BICAirline

Ithuteng Ntakhwana ID: 201801847

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```
#Load Air passengers dataset embedded in R
passengers = data.frame(year = rep(1949:1960, each = 12),
                        month = c(1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12),
                        fq = c(112, 118, 132, 129, 121, 135, 148, 148, 136, 119, 104, 118,
                               115, 126, 141, 135, 125, 149, 170, 170, 158, 133, 114, 140,
                               145, 150, 178, 163, 172, 178, 199, 199, 184, 162, 146, 166,
                               171, 180, 193, 181, 183, 218, 230, 242, 209, 191, 172, 194,
                               196, 196, 236, 235, 229, 243, 264, 272, 237, 211, 180, 201,
                               204, 188, 235, 227, 234, 264, 302, 293, 259, 229, 203, 229,
                               242, 233, 267, 269, 270, 315, 364, 347, 312, 274, 237, 278,
                               284, 277, 317, 313, 318, 374, 413, 405, 355, 306, 271, 306,
                               315, 301, 356, 348, 355, 422, 465, 467, 404, 347, 305, 336,
                               340, 318, 362, 348, 363, 435, 491, 505, 404, 359, 310, 337,
                               360, 342, 406, 396, 420, 472, 548, 559, 463, 407, 362, 405,
                               417, 391, 419, 461, 472, 535, 622, 606, 508, 461, 390, 432
                        ))
#Time Series Object
passengers_time = ts(passengers[,-c(1,2)], frequency = 12,start = c(1949,1))
passengers_time
```

```
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
```

```
#Variables of interest to store information
best_or = NULL
best_model = NULL
best_bic = Inf
```

```r
#loops in p,d,q ARIMA model
for(p in 0:3) {
  for (d in 0:1) {
    for (q in 0:3) {
      #Fitting ARIMA(p,d,q) model with Max. Likelihood
      a_model = tryCatch(arima(passengers_time, order = c(p,d,q), method = "ML"),
                         error = function(e) NULL
                         )
      #Checking if model had fit successfully
      if (!is.null(a_model)) {
        bicmod = BIC(a_model)

        #Updating model if BIC is lower than best fit
        if (bicmod < best_bic) {
          best_or = c(p,d,q)
          best_model = a_model
          best_bic = bicmod
        }
      }
    }
  }
}
```

```
## Warning in arima(passengers_time, order = c(p, d, q), method = "ML"): possible
## convergence problem: optim gave code = 1
```

```r
#Print of the best model
cat("The best order model is:", best_or,"\n")
```

```
## The best order model is: 3 1 3
```

```r
print(best_model)
```

```
##
## Call:
## arima(x = passengers_time, order = c(p, d, q), method = "ML")
##
## Coefficients:
##          ar1     ar2      ar3      ma1      ma2     ma3
##       1.0085  0.1453  -0.5534  -1.0158  -0.7027  0.8924
## s.e.  0.1043  0.1727   0.1041   0.0720   0.1179  0.0704
##
## sigma^2 estimated as 605.2:  log likelihood = -665.19,  aic = 1344.38
```

```r
cat("BIC:", best_bic, "\n")
```

```
## BIC: 1365.125
```